

Assignment - 9

- ① Derive the law $\partial_\mu F^{\mu\nu} = \mu_0 J^\nu$ from the Maxwell's action.

Ans: $S_{EM}[A^\mu] = \int d^4x \left(-\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} + J_\mu A^\mu \right)$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

From the least action principle

$$\delta S_{EM} = 0$$

$$\delta S_{EM} = \int d^4x \left[-\frac{1}{4\mu_0} \delta (F_{\mu\nu} F^{\mu\nu}) - \delta (J_\mu A^\mu) \right]$$

① $\delta (F_{\mu\nu} F^{\mu\nu})$

$$\underbrace{F_{\mu\nu} \delta (F^{\mu\nu})}_{(a)} + F^{\mu\nu} \delta (F_{\mu\nu})$$

$$\textcircled{a} = F_{\mu\nu} \delta F^{\mu\nu} = \eta^{\mu\alpha} \eta^{\nu\beta} F_{\alpha\beta} \delta F_{\mu\nu}$$

$$\rightarrow \delta F^{\mu\nu} = \eta^{\mu\alpha} \eta^{\nu\beta} \delta F_{\alpha\beta}$$

$$\rightarrow F_{\mu\nu} \delta F^{\mu\nu} = \left\{ \eta^{\mu\alpha} \eta^{\nu\beta} F_{\mu\nu} \right\} \delta F_{\alpha\beta} = F^{\alpha\beta} \delta F_{\alpha\beta}$$

$$\rightarrow \boxed{F_{\mu\nu} \delta F^{\mu\nu} = 2 F^{\mu\nu} \delta F_{\mu\nu}}$$

$$\therefore \textcircled{1} = \boxed{\delta (F_{\mu\nu} F^{\mu\nu}) = 2 F^{\mu\nu} \delta F_{\mu\nu}} \quad \textcircled{1}$$

$$\therefore \delta S_{EM} = \frac{-1}{2\mu_0} \int d^4x \underbrace{F^{\mu\nu} \delta F_{\mu\nu}}_{\textcircled{1}} + \int d^4x J_M^\mu \delta (A_\mu)$$

$$\textcircled{1} = F^{\mu\nu} \delta (\partial_\mu A_\nu - \partial_\nu A_\mu)$$

$$= F^{\mu\nu} [\partial_\mu (\delta A_\nu) - \partial_\nu (\delta A_\mu)]$$

$$= \underbrace{F^{\mu\nu} \partial_\mu (\delta A_\nu)}_{\textcircled{ii}} - F^{\mu\nu} \partial_\nu (\delta A_\mu) \quad \textcircled{1}$$

Relabelling indices in $\textcircled{iii}$:

$$F^{\mu\nu} \partial_\mu (\delta A_\nu) = F^{\nu\mu} \partial_\nu (\delta A_\mu)$$

$$= - F^{\mu\nu} \partial_\nu (\delta A_\mu) = \textcircled{iv}$$

Substituting $\textcircled{iv}$:

$$\textcircled{1} = + 2 F^{\mu\nu} \partial_\mu (\delta A_\nu)$$

$$\therefore \delta S_{EM} = -\frac{1}{\mu_0} \int d^4x F^{\mu\nu} \partial_\mu (\delta A_\nu) - \int d^4x J_M^\mu \delta (A_\mu)$$

(ignoring boundary term)

(integrating by parts)

$$= -\frac{1}{\mu_0} \int \partial_\mu F^{\mu\nu} \delta A_\nu - \int d^4x J_M^\mu \delta A_\mu$$

$$= \int \frac{1}{\mu_0} \left[\frac{1}{\mu_0} \partial_\mu F^{\mu\nu} + J_M^\nu \right] \delta A_\nu d^4x = 0$$

$$\left[J_M^\nu \delta A^\mu = J_M^\mu \delta A^\nu \right]$$

$\delta S_{EM} [A^\mu] = 0$ for all A^μ only when:

$$\frac{1}{\mu_0} \partial_\mu F^{\mu\nu} + J^\nu = 0 \quad (13.10)$$

$$\rightarrow \partial_\mu F^{\mu\nu} = -\mu_0 J^\nu \quad (\text{Proved})$$

(2) Evaluate $I_1 = \int_0^\pi \sin \theta e^{ikr \cos \theta} d\theta$.

Ans: Let $x = \cos \theta$

$$\rightarrow dx = -\sin \theta d\theta$$

@ $\theta = 0 \rightarrow \cos \theta = 1 = x_i$

@ $\theta = \pi \rightarrow \cos \theta = -1 = x_f$

$$\therefore I_1 = - \int_{-1}^1 dx e^{ikrx}$$

$$= \int_{-1}^1 e^{ikrx} dx$$

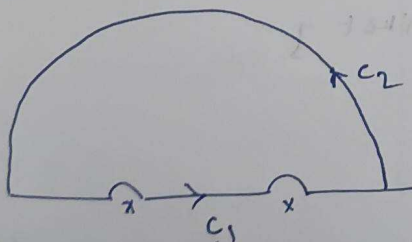
$$= \left[\frac{e^{ikrx}}{ikr} \right]_{-1}^1$$

$$= \frac{e^{ikr} - e^{-ikr}}{ikr} = \frac{2 \sin(kr)}{kr}$$

$$\therefore I_1 = \frac{2 \sin(kr)}{kr} \quad (\text{Proved Ans})$$

(3) Calculate $I_2 = \int_{-\infty}^{\infty} \frac{d\omega}{(\omega - ck)(\omega + ck)} e^{-i\omega t}$

Ans: For $t < 0$:



$$C = C_1 + C_2$$

$$\therefore I_2 = \int_{C_1} \left(\frac{d\omega}{(\omega - ck)(\omega + ck)} e^{-i\omega t} \right)$$

$$= \int_{C_1} \left(\frac{d\omega}{(\omega - ck)(\omega + ck)} e^{-i\omega t} \right) - \int_{C_2} \left(\frac{d\omega}{(\omega - ck)(\omega + ck)} e^{-i\omega t} \right)$$

C does not contain any poles singular points inside it.

Hence, from Cauchy's Theorem:

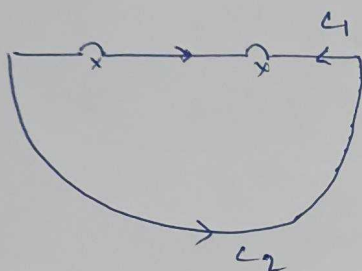
$$\oint_C \frac{dw e^{-iwt}}{(w-ck)(w+ck)} = 0$$

For calculation of the contour C_2 , we use Jordan's Lemma. Since the integrand $\rightarrow 0$ as $w \rightarrow \infty$, hence

$$\int_{C_2} \frac{dw e^{-iwt}}{(w-ck)(w+ck)} = 0$$

$$\therefore \boxed{I_2 = 0} \text{ for } t < 0$$

For $t > 0$:



From Jordan's Lemma,

$$\int_{C_2} \frac{e^{-iwt}}{(w-ck)(w+ck)} dw = 0$$

From Cauchy's Theorem,

$$\oint_C f(z) dz = \pm 2\pi i \sum \text{Res } f(z) |_{z=z_0}$$

There are 2 poles in C @ $w = ck$ & $w = -ck$ of order 1.

$$\begin{aligned} &= 2\pi i \left[\frac{e^{-iwt}}{(w-ck)(w+ck)} \Big|_{w=ck} + \frac{e^{-iwt}}{(w-ck)(w+ck)} \Big|_{w=-ck} \right] \\ &= \frac{2\pi i}{ck} \left[\frac{e^{-ickt}}{2ck} - \frac{e^{ickt}}{2ck} \right] \\ &= -\frac{\pi i}{ck} [e^{ickt} - e^{-ickt}] \\ &= -\frac{\pi i}{ck} (2i \sin(ckt)) \\ &= \frac{2\pi}{ck} \sin(ckt) \end{aligned}$$

for $t > 0$ (Ans)

$$\therefore I_2 = \frac{2\pi}{ck} \sin(ckt)$$

$$\boxed{\frac{2\pi}{k} \sin(ckt) = I_2}$$